

Library Management System

Python Essentials Course Project Report

Course: Python Essentials / Problem Solving

Interface: Command Line (CLI)

1. Project Overview

For this course project, I developed a simple and interactive command-line Library Management System using Python. The primary goal of the application is to streamline core administrative operations within a library environment. This includes managing book entries, displaying real-time stock availability, tracking borrowing transactions, and processing book returns alongside automatic fee calculations for overdue items.

Building this program as a CLI application provided a solid practical foundation in essential Python programming constructs, such as dynamic control flow, user input validation, data state maintenance, and logical conditionals—all without the overhead of external graphical framework dependencies.

2. Key Features Implemented

- **1. Add Book:** Collects book details (serial number and title) to create new inventory records.
- **2. View Books:** Formats and prints the complete inventory list, explicitly showing each book's current status (Available or Issued).
- **3. Search Book:** Enables fast lookups via serial numbers to immediately display matching book titles and borrowing statuses.
- **4. Issue Book:** Updates the availability status from Available to Issued upon loaning a volume.
- **5. Return Book:** Validates return requests, calculates overdue late fees based on loan duration, and updates book status back to Available.
- **6. Exit:** Terminates session execution gracefully with a standard closing confirmation prompt.

3. Program Workflow & Execution Analysis

Below is a summary of the program's operations during the execution test session:

Serial No.	Book Title	Initial Status	Action Taken	Final Status
101	Python	Available	Added to inventory	Available
102	C++	Available	Issued & Returned (23 Days)	Available

Serial No.	Book Title	Initial Status	Action Taken	Final Status
103	Java	Available	Searched & Verified	Available

Return Logic & Fine Calculation Test:

During the execution run, returning Book 102 after 23 days triggered the fine evaluation logic, calculating a fine of ₹160 before successfully marking the item as returned. Unissued return attempts (such as targeting Book 103 prior to borrowing) correctly triggered validation safeguards: *"This book is not issued."*

4. Key Technical Learnings

- **Input Validation & Guard Clauses:** Handled invalid serial numbers and non-borrowed return attempts using robust conditional flow control.
- **State Management:** Maintained in-memory state changes during terminal execution to dynamically update book availability statuses.
- **Fine Evaluation Logic:** Implemented mathematical calculations to automatically compute variable late fines based on total usage days.

5. Instructions to Run the Application

To run the program locally on your machine, follow these simple steps:

1. Ensure **Python 3.x** is installed on your operating system.
2. Open your command line interface (Terminal or Command Prompt) and navigate to the project root directory.
3. Execute the program file using:

```
python code.py
```

Use options 1 through 6 in the main menu prompt to navigate and operate the library system.